PERMUTATIONS & COMBINATIONS: ORDER MATTERS!

Lesson Guide — Secret Locks, Podium Lineups & Ice Cream Bowls

1. The Core Concept: Arranging vs. Choosing

In daily life, we constantly group and organize items. However, in mathematics, there is a fundamental difference between **arranging items where order matters** and **choosing items where order does not matter**.

1. The Great Lock Mystery

Consider a combination lock with digits. If the secret code to open a treasure chest is $9 - 7 - 4$, will typing $4 - 7 - 9$ unlock it?

No! Even though both codes use the exact same digits ($4, 7, 9$), the sequence matters. A secret lock requires a specific *Permutation*.

2. Key Distinctions

Category	ARRANGING (PERMUTATION)	CHOOSING (COMBINATION)
Does Order Matter?	**YES**	**NO**
Key Action	Line up, order, rank, sequence, or position.	Select, group, pick, combine, or sample.
Real-World Examples	• Winner's podium (1st, 2nd, 3rd) • Seating arrangements on chairs • Lock codes & passwords • School assembly line-ups	• Scoops of ice cream in a bowl • Selecting a cricket team from a squad • Fruit salad ingredients • Carrom team pairs

3. Counting Arrangements: The Slot Method

When order matters, we use the **Slot Method** (Fundamental Counting Principle) to find the total number of unique arrangements.

Seating Challenge: Ananya, Rohan & Diya

Suppose we have 3 chairs (Chair 1, Chair 2, Chair 3) and 3 friends to seat:

- **Slot 1 (1st Chair):** 3 potential choices (Ananya, Rohan, or Diya).
- **Slot 2 (2nd Chair):** 2 remaining choices.
- **Slot 3 (3rd Chair):** 1 remaining choice.

Total Arrangements = $3 \times 2 \times 1 = 6$ distinct ways

4. Counting Selection: Combinations

When order does *not* matter, swapping the position of chosen items produces the exact same group.

Example A: Ice Cream Bowl (2 Scoops from 3 Flavors)

Flavors available: **Mango, Kulfi, Chocolate**. How many 2-scoop bowls can you make?

- Pair 1: Mango + Kulfi
- Pair 2: Kulfi + Chocolate
- Pair 3: Mango + Chocolate

Note: Mango + Kulfi is identical to Kulfi + Mango because both sit in the same bowl! Total = 3 combinations.

Example B: Sweet Shop Helper (2 Sweets from 3 Types)

Sweets available: **Jalebi, Laddu, Barfi**. How many unique 2-sweet gift boxes can you pick?

- Box 1: Jalebi & Laddu
- Box 2: Laddu & Barfi
- Box 3: Jalebi & Barfi

Total = 3 unique combinations.

5. Lesson Summary & Recap

- **Order Matters:** When arranging items, forming passwords, or assigning positions (Permutations).
- **Order Doesn't Matter:** When selecting items into a group or collection (Combinations).
- **Slot Method:** Multiply the number of available options for each position to count total arrangements.
- **No Repeats & No Order:** Pure combinations focus solely on which items are included in the group.

STUDENT PRACTICE WORKSHEET

Permutations & Combinations — Test Your Knowledge

Instructions: Answer the following questions based on the concepts of arranging vs. choosing. (Do not write answers on this sheet).

Question 1 (Multiple Choice)

In which of the following scenarios does **ORDER MATTER**?

- A) Choosing 3 fruits to make a smoothie
- B) Setting a 4-digit PIN code for a mobile phone
- C) Selecting 2 books from a shelf to read over the weekend
- D) Picking a committee of 4 students from a class

Question 2 (Multiple Choice)

A sweet shop offers Jalebi, Laddu, Gulab Jamun, and Barfi. You want to pick a box of 2 DIFFERENT sweets. How many unique sweet box combinations can you make?

- A) 4
- B) 6
- C) 12
- D) 24

Question 3 (Slot Method Computation)

There are 4 runners competing in a 100m sprint. In how many different ways can Gold, Silver, and Bronze medals (1st, 2nd, and 3rd place) be awarded?

- A) 12 ways
- B) 24 ways
- C) 6 ways
- D) 4 ways

Question 4 (Categorization)

Classify each of the following four real-world activities as either a **Permutation (Arrangement)** or a **Combination (Selection)**:

1. Assigning 1st, 2nd, and 3rd rank prizes in a math competition.
2. Selecting 11 players for a cricket match out of a 15-player squad.
3. Arranging 5 distinct textbooks side-by-side on a bookshelf.
4. Picking 2 toppings for a pizza from 5 available choices.

Question 5 (Short Answer & Explanation)

Explain why a 3-digit combination lock code (e.g., $8 - 2 - 5$) is mathematically a *Permutation* rather than a *Combination*.

Question 6 (Daily Micro-Challenge)

Write down one original real-life example from your daily routine where **ORDER MATTERS** and one example where **ORDER DOES NOT MATTER**.